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# Data Engineering Project- Airline on-time performance analysis

CIS 9760- BIG DATA TECHNOLOGIES

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Milestone 1

**Proposal for Airline Data Analysis**

**Dataset Description**
The dataset contains records of airline performance, including details on flight schedules, delays, cancellations, and other operational metrics. This dataset can be leveraged to analyze trends in airline efficiency, punctuality, and factors influencing delays and cancellations.

**Data Source**
The dataset is available for download at the following location: Airline on-time performance analysis

**Dataset Attributes**
The dataset includes 29 columns that capture relevant flight details. Attributes are:

1. **Year** – 1987-2008
2. **Month** – 1-12
3. **DayofMonth** – 1-31
4. **DayofWeek** – 1(Monday) – 7(Sunday)
5. **DepTime** – actual depature time (local, hhmm)
6. **CRSDepTime** – scheduled departure time (local, hhmm)
7. **ArrTime** – actual arrival time (local, hhmm)
8. **CRSArrTime** – scheduled arrival time (local,hhmm)
9. **UniqueCarrier** – unique carrier code
10. **Flight Number** – flight number
11. **Tailnum** – plane tail number
12. **ActualElapsedTime** – in minutes
13. **CRSElapsedTime** – in minutes
14. **AirTime** – in minutes
15. **ArrDelay** – arrival delay, in minutes
16. **DepDelay** – departure delay, in minutes
17. **Origin** – origin IATA airport code
18. **Dest** – destination IATA airport code
19. **Distance** – in miles
20. **TaxiIn** – tax in time, in minutes
21. **TaxiOut** – taxi out time in minutes
22. **Cancelled** – was the flight cancelled
23. **CancellationCode** – reason for cancellation (A = carrier, B = weather, C = NAS, D = security)
24. **Diverted** – 1 = yes, 0 = no
25. **CarrierDelay** – in minutes
26. **WeatherDelay** – in minutes
27. **NasDelay** – in minutes
28. **SecurityDelay** – in minutes
29. **LateAircraftDelay** – in minutes

**Prediction Objective:**

The goal of this analysis is to predict flight delays using machine learning techniques. Based on the nature of the data and the objectives, **logistic regression** will be used to classify flights as "On Time" or "Delayed" based on whether the delay exceeds a certain percentage (X%) of the total trip time. This method is suitable for categorizing flights effectively while allowing for different delay thresholds (e.g., 10%, 25%, 50%).

**Machine Learning Approach**

By implement logistic regression to predict whether a delay exceeds a given percentage of total trip time. Additionally, we will experiment with different threshold values (10%, 25%, and 50%) to assess how they impact classification accuracy. Other models such as decision trees or random forests may also be explored for improved performance.

In conclusion, this analysis will provide valuable insights into flight delays, helping airlines and passengers make more informed travel decisions. By identifying key contributing factors to develop better strategies to minimize disruptions in air travel.